

Hao Pan

AI Application Engineer | Agent Workflows & Data Applications

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Summary

Master of Computing candidate in Applied AI at NTU building Agent workflows and data applications with Python, TypeScript, SQL, and AI coding tools such as Claude Code. At Shopee, develops reusable A/B Agent Skills and has delivered an LLM-assisted knowledge-base MVP with controlled routing, evidence-bound outputs, and human approval; also built a public BI Data Agent with read-only SQL, guardrails, tests, and CI.

Education

Nanyang Technological University

Master of Computing in Applied Artificial Intelligence, College of Computing and Data Science

Nov 2025 - Jan 2027 (Expected)

Singapore

Suzhou University

B.Eng. in Data Science and Big Data Technology; National Scholarship (top 0.2% nationwide)

Sep 2019 - Jun 2023

China

Experience

Shopee Singapore

AI Data Analyst Intern, Buyer & Marketing Business Intelligence

Mar 2026 - Present (through Sep 2026)

Singapore

A/B Analysis Agent Skills (Internal Agent platform; used internally)

- Packaged validated business SQL, experiment parameters, metric definitions, and acceptance criteria into reusable Agent Skills with defined inputs, outputs, parameter checks, and pre-return checks; iterated templates and User Guides with cross-country BI teams and platform product managers.
- Reconciled key metrics against baselines under matched inputs; when a frozen query repeatedly timed out, retained both execution traces and stopped after the predefined retry limit rather than returning a partial result.

Analysis Agent Copilot (LLM-assisted application built on a team case knowledge base; internal testing with real historical cases)

- Built a Python ingestion and validation pipeline for a 169-item executable import plan; curated 17 traceable, review-ready cases with source and version records, with current structure, reference, and version checks at 0 errors and 0 warnings.
- Combined live Google Sheets values with XLSX layout in a version-bound data path, preserving source errors and recovering 170 values missing from exported files in a real pilot.
- Implemented the Svelte frontend and a layered Python backend; integrated the platform LLM for intent handling and candidate explanation, with routing limited to approved handlers, analyst selection required, and structured drafts generated only from the selected evidence and fixed version.
- Protected high-risk ingestion with backend allowlists, validation, and human approval; verified the review-branch, commit, and merge-request path without writing directly to main.

Jinken Vocational and Technical College

Lecturer & AI Technology Lead

Mar 2023 - Oct 2025

China

- Designed and delivered Python, machine learning, and generative AI courses to 300+ students, translating technical concepts into labs, assessments, and project work.
- Led a ROS-based indoor autonomous-flight and obstacle-avoidance project; coached the team to Third Prize at the 2024 RAICOM National Finals.

Selected Projects

Public BI Data Agent (Deterministic, Synthetic Data)

TypeScript, React, Node/Vercel, Zod, AlaSQL, Vitest

Jul 2026

Live Demo ▪ GitHub

- Built and deployed a public BI data-agent demo: deterministic rules map supported questions to metrics, tables, and read-only SQL over isolated in-memory synthetic data, returning SQL, results, charts, and reviewable traces.
- Added request-schema, known-table and column, read-only SQL, and sensitive-field checks; blocks user-level exports before execution and suggests aggregate alternatives. CI on the public main branch passes typecheck, lint, build, and 172 automated tests across request boundaries, execution-state/result consistency, and NDJSON stream completeness/event ordering.

Technical Skills

Development: Python, TypeScript, Svelte, FastAPI, Node.js, React / Next.js, REST APIs, SQL, Zod, Vitest, Git / GitLab, Linux

AI Application Engineering: Platform LLM integration, controlled handler routing, structured outputs, evidence constraints, human approval, guardrails, automated testing, failure handling

Data Platforms: Presto / Trino, Hive, PySpark, pandas, Google Drive / Sheets APIs

Languages & Global: Chinese (native); English (IELTS 6.5), used in cross-country BI requirement alignment in Singapore

Honors: Huawei ICT Competition Global Final, Cloud Track, Third Prize (top 5%, Jun 2022)